SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1507

COURSE OUTCOME COVERED
CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

Code of Conduct
I Dinesh S (192521152) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Dinesh S

Assessment Tasks

Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices.Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. **Case Study 3: Online Food Ordering Platform**

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients.
Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.
Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory management system to connect all its stores and manage products from a centralized platform. The system collects information from POS machines, barcode scanners, warehouse systems, and store dashboards. This information is sent to cloud services through secure APIs.

Additional Content

- Centralized Management: All store inventory information is maintained in one cloud-based system.
- Real-Time Stock Updates: When a product is sold, the inventory quantity is updated immediately.
- Product Tracking: Each product can be tracked using product IDs or barcodes.
- Sales Data: The system records daily sales, returns, and purchase transactions.
- Warehouse Management: Warehouse stock can be monitored along with individual store stock.
- Demand Prediction: Analytics services study previous sales to predict future product demand.
- Automatic Alerts: The system can notify managers when stock reaches a low level.
- Data Backup: Important inventory and transaction data can be backed up in cloud storage.
- Load Balancing: Cloud infrastructure distributes user requests across multiple servers.
- High Availability: Backup servers and cloud resources help keep the system available.
- Scalability: More computing resources can be added when the number of stores increases.
- Data Analytics: Managers can analyze sales trends and identify popular products.
- Cost Efficiency: The company can reduce the need for maintaining large physical servers.
- Disaster Recovery: Cloud backups help recover data after hardware failures or other problems.
- Security: Authentication, firewalls, VPNs, and API gateways prevent unauthorized access.

Overall Benefit

The system helps the retail company reduce stock problems, improve sales forecasting, manage multiple stores, reduce operational costs, and make better business decisions.

Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a cloud-based document collaboration platform where employees can create, edit, store, and share documents online. Users can access documents from different locations and devices without installing large software applications locally.

Additional Content

- Real-Time Editing: Multiple users can work on the same document.
- Document Sharing: Users can share documents with selected people or teams.
- Version History: Previous document versions are maintained for recovery and tracking.
- Automatic Saving: Changes can be automatically saved to cloud storage.
- Multi-Device Access: Users can access documents from computers, tablets, and smartphones.
- Synchronization: Changes made on one device are synchronized with other devices.
- User Management: Administrators can create and manage user accounts.
- Access Control: Different users can receive different permissions.
- Comments: Team members can add comments and suggestions to documents.
- Cloud Database: Stores user information, document metadata, and activity records.
- Distributed Storage: Documents can be stored across multiple cloud servers.
- Backup: Documents can be backed up to prevent data loss.
- High Availability: Cloud infrastructure allows users to access documents continuously.
- Network Optimization: Cloud servers and content delivery infrastructure can reduce access delays.
- Security: Encryption protects documents while they are being transmitted and stored.
- Monitoring: Administrators can monitor system performance and user activities.

Overall Benefit

The system provides easy document access, real-time teamwork, secure storage, automatic synchronization, version management, and improved productivity.

Case Study 3: Online Food Ordering Platform

A startup develops a cloud-based food ordering platform that allows customers to browse restaurants, select food items, place orders, and track deliveries. The platform uses web and mobile applications connected to cloud-based backend services.

Additional Content

- Customer Registration: Customers can create and manage their accounts.
- Restaurant Management: Restaurants can add menus, prices, images, and availability.
- Food Search: Customers can search for restaurants and food items.
- Shopping Cart: Selected food items can be added to a digital cart.
- Order Management: Backend services process and track customer orders.
- Order Status: Customers can view whether an order is accepted, prepared, or delivered.
- Cloud Database: Stores customer, restaurant, menu, and order information.
- Cloud Storage: Stores food images, invoices, and other files.
- REST APIs: Enable communication between mobile/web clients and backend services.
- Virtual Machines: Provide computing resources for application services.
- Kubernetes: Helps manage containers and scale services automatically.
- Load Balancing: Distributes customer requests across multiple servers.
- Scalability: Cloud resources can increase during peak ordering periods.
- Security: HTTPS protects data transferred between clients and servers.
- Authentication Tokens: Verify users when they access protected services.
- Role-Based Access: Customers, restaurants, delivery staff, and administrators have different permissions.
- Monitoring: Cloud monitoring helps identify performance problems and failures.
- Backup and Recovery: Important customer and order data can be backed up.
- High Availability: Multiple cloud resources help keep the service running.
- Analytics: The company can analyze orders, popular food items, and customer activity.

Overall Benefit

The platform provides fast online ordering, secure data management, scalability, reliable service, real-time order processing, and improved customer experience.